

Software Engineering Lab Assignment-3

World In Brief — AI News Intelligence Platform

System Design Documentation: DFDs, Sequence & Collaboration Diagrams

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1. Introduction

World In Brief is an AI-powered News Intelligence Platform designed to aggregate, process, and deliver personalized news content to users. The system leverages artificial intelligence (specifically Ollama AI with Llama 3.2) to summarize articles, classify content, and generate intelligent responses to user queries. The platform supports multi-channel delivery including web browsers, mobile clients, and push notifications via email/SMS gateways.

This document presents a comprehensive system design using **Data Flow Diagrams (DFDs)** at multiple levels of abstraction (Level 0, Level 1, and Level 2), along with **UML Behavioral Diagrams** (Sequence Diagram and Collaboration Diagram) to illustrate the dynamic interactions between system components.

The DFD notation follows standard conventions where:

Symbol	Shape	Description
Process	Rounded Rectangle	Transforms incoming data into outgoing data
External Entity	Rectangle	Sources/destinations of data outside system boundary
Data Store	Open-ended Rectangle	Repositories where data is stored
Data Flow	Arrow	Direction of data movement between components

The system architecture follows a modular design principle, separating concerns into distinct processes such as user authentication, news ingestion, AI processing, personalized feed generation, notification management, and administrative functions. This document explains each diagram in detail, highlighting data flows, process responsibilities, and interactions with external entities.

2. Data Flow Diagrams (DFDs)

Data Flow Diagrams are graphical representations that illustrate how data moves through a system, showing the processes that transform data, the data stores where information is held, the external entities that interact with the system, and the data flows that connect these elements. The following sections present DFDs at three levels of abstraction, from the high-level context view to detailed process decompositions.

2.1 DFD Level 0 — Context Diagram

The **Level-0 DFD (Context Diagram)** provides the highest level of abstraction, representing the entire **World In Brief** system as a single process. This diagram establishes the system boundary and identifies all external entities that interact with the platform, along with the major data flows between them.

Central Process — "World In Brief System": Represented as a single rounded rectangle, this encapsulates the entire AI News Intelligence Platform. All internal complexities are hidden at this level, providing a clear black-box view of the system.

External Entities:

- **User:** The primary consumer of news content. Users submit search queries, view articles, provide feedback, and manage personal preferences. Data flows include search requests, article views, feedback submissions, and personalized content delivery.
- **Admin:** System administrators responsible for content moderation, user management, configuration, and analytics monitoring. Admins receive moderation flags, analytics dashboards, and send configuration updates.
- **External News API / RSS:** Third-party news sources and RSS feeds that provide raw news articles, metadata, and updates. The system sends API requests and RSS fetch commands, receiving parsed articles and metadata in return.
- **Ollama AI (Llama 3.2):** The external AI service used for content summarization, classification, and intelligent query responses. The system sends article content and user queries, receiving AI-generated summaries, classifications, and responses.
- **Browser / Mobile Client:** The presentation layer through which users interact with the system. Handles UI rendering, user actions, and notification delivery.
- **Email/SMS Gateway:** External communication service for delivering push notifications, breaking news alerts, and digest emails to users.

Significance: The Context Diagram is essential for stakeholders to understand the system's scope, its relationship with the external environment, and the primary data exchanges without being overwhelmed by internal details. It serves as a communication tool between technical teams and business stakeholders.

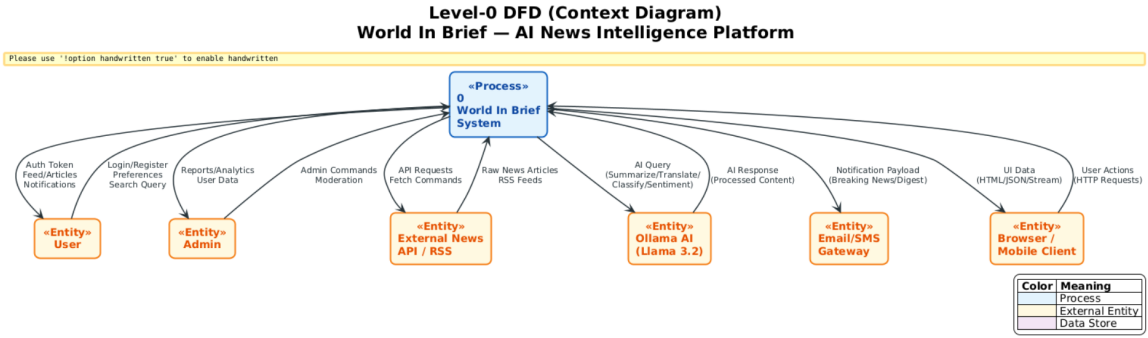


Figure 1: Level-0 DFD (Context Diagram) — World In Brief AI News Intelligence Platform

2.2 DFD Level 1 — System Decomposition

The **Level-1 DFD** decomposes the single process from the Context Diagram into eight major sub-processes, revealing the internal architecture of the **World In Brief** platform. This diagram shows how data flows between these processes, data stores, and external entities, providing a comprehensive view of the system's functional decomposition.

Major Processes (Level-1):

Process 1.0 — Authentication & User Management: Handles user registration, login/logout, profile management, and password reset operations. Interacts with the User DB (D1) and Session Cache (D9) to maintain authentication state. Receives credentials from users and validates them against stored records.

Process 2.0 — News Ingestion & Management: Responsible for fetching news from external APIs and RSS feeds, parsing content, classifying articles, validating data quality, indexing for search, and detecting duplicates. Stores processed articles in the News Article DB (D2) and maintains trending data in the Trending DB (D6).

Process 3.0 — Personalized Feed Engine: Generates customized news feeds for each user based on preferences, reading history, bookmarks, and AI-driven recommendations. Interacts with multiple data stores including User Preferences (D3), Reading History (D5), News Articles (D2), and Bookmarks (D4).

Process 4.0 — AI Intelligence Processor: The core AI module that leverages Ollama AI (Llama 3.2) for content summarization, topic extraction, sentiment analysis, and intelligent query processing. Manages AI response caching to optimize performance.

Process 5.0 — Trending & Analytics: Analyzes article engagement metrics, extracts trending topics, calculates engagement scores, and generates analytics dashboards for administrators.

Process 6.0 — Notification Manager: Manages user notification preferences, detects breaking news, compiles daily/weekly digests, and dispatches notifications through email/SMS gateways.

Process 7.0 — Bookmark & Offline Manager: Allows users to bookmark articles for later reading, manages offline content caching, and synchronizes bookmark data across devices.

Process 8.0 — Feedback & Admin Console: Collects user feedback and ratings, provides content moderation tools, handles bug reports, and offers administrative dashboards with analytics.

Data Stores at Level-1:

- **D1 — User DB:** Stores user profiles, credentials, and account information.
- **D2 — News Article DB:** Repository of all ingested and processed news articles.
- **D3 — Preferences DB:** User-specific settings and content preferences.

- **D4 — Bookmark DB:** User bookmarked articles and reading lists.
- **D5 — Reading History:** Tracks user reading patterns and article interactions.
- **D6 — Trending DB:** Aggregated trending topics and popularity metrics.
- **D7 — Notification DB:** Notification logs, schedules, and delivery status.
- **D8 — Admin Logs:** Administrative actions and system audit trails.
- **D9 — Session Cache:** Active user sessions and authentication tokens.
- **D10 — Feedback DB:** User ratings, comments, and feedback submissions.

Significance: Level-1 DFD provides the architectural blueprint of the system, showing how responsibilities are distributed across modules. It identifies key data stores and establishes the foundation for further decomposition in Level-2 DFDs.

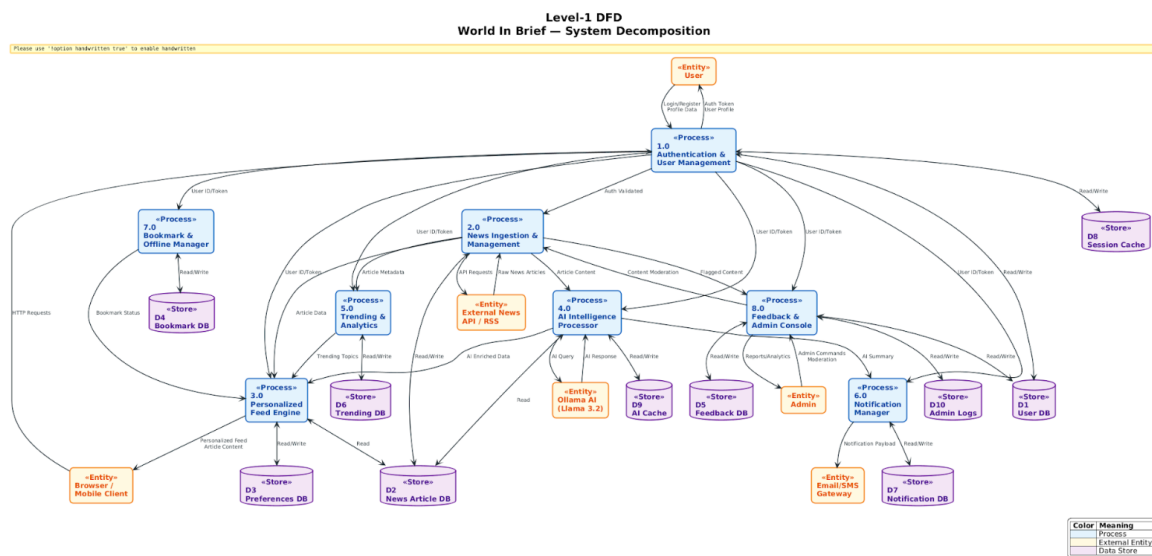


Figure 2: Level-1 DFD — System Decomposition of World In Brief Platform

2.3 DFD Level 2 — Process Decomposition

The **Level-2 DFDs** provide the most detailed view, exploding each Level-1 process into its constituent sub-processes. This section presents the decomposition of all eight major processes, revealing the granular data transformations, validation logic, and storage operations within each module.

2.3.1 Level-2 DFD — Process 2.0: News Ingestion & Management

This diagram details how raw news is transformed into structured, searchable content through a pipeline of specialized sub-processes:

- **Process 2.1 — News Fetcher & RSS Parser:** Retrieves articles from external APIs and RSS feeds. Sends API requests with parameters like category, date range, and source filters. Receives raw article data including title, content, author, publication date, and metadata.
- **Process 2.2 — Content Classifier:** Categorizes articles by topic (politics, technology, sports, etc.), genre (news, opinion, analysis), and relevance score. Uses both rule-based and ML-based classification techniques.
- **Process 2.3 — Article Validator:** Validates content quality by checking for completeness (title, body, author, source), detecting spam or low-quality content, and verifying source credibility. Flags suspicious articles for admin review.
- **Process 2.4 — Article Indexer:** Creates inverted search indexes, extracts keywords, generates metadata tags, and builds full-text search capabilities for fast article retrieval.
- **Process 2.5 — Duplicate Detector:** Identifies and handles duplicate or near-duplicate articles using fuzzy matching and hash-based comparison. Prevents content redundancy in the database.
- **Data Stores:** D2 (News Article DB) for persistent storage, D6 (Trending DB) for popularity tracking.
- **External Entity:** Admin receives moderation flags and validation reports for manual review.

Level-2 DFD Process 2.0 — News Ingestion & Management

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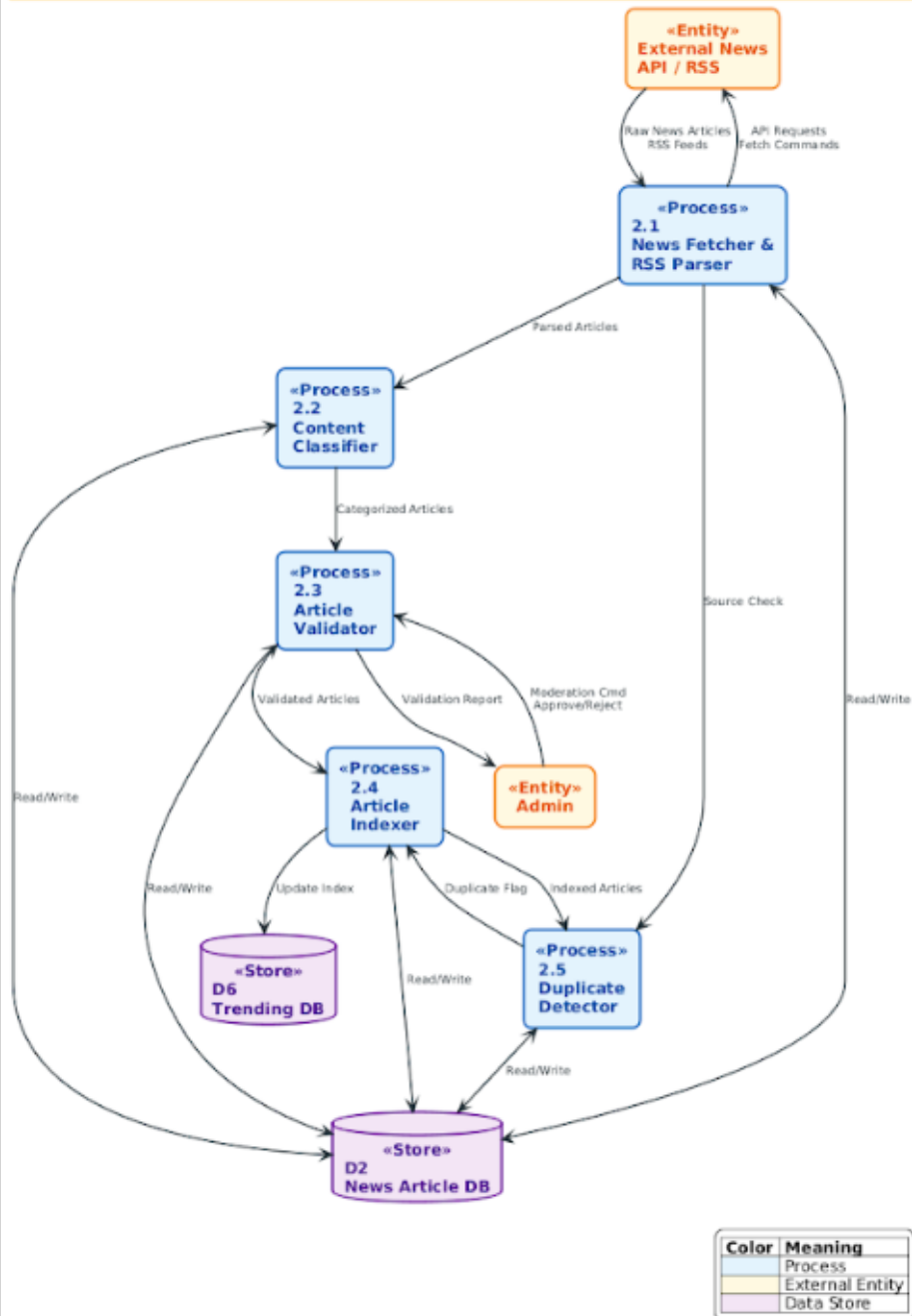


Figure 3: Level-2 DFD — Process 2.0: News Ingestion & Management

2.3.2 Level-2 DFD — Process 1.0: Authentication & User Management

This diagram shows the complete user identity lifecycle from registration to account recovery:

- **Process 1.1 — Register User:** Creates new accounts with input validation (email format, password strength, username uniqueness). Stores initial profile data in User DB (D1).
- **Process 1.2 — Email Validation:** Verifies email addresses through confirmation links sent via Email/SMS Gateway. Updates validation status in User DB.
- **Process 1.3 — Login & JWT Issuance:** Authenticates users against stored credentials, generates JSON Web Tokens (JWT) for session management, and stores active sessions in Session Cache (D9). Supports multi-factor authentication.
- **Process 1.4 — Profile Management:** Allows users to update personal information, preferences, and privacy settings. Validates changes before persisting to User DB.
- **Process 1.5 — Password Reset:** Handles secure password recovery via email verification. Generates temporary tokens, validates reset requests, and updates credentials in User DB.
- **Data Stores:** D1 (User DB) for persistent user data, D9 (Session Cache) for active sessions.
- **External Entity:** Mobile Client receives push notifications for account-related events.

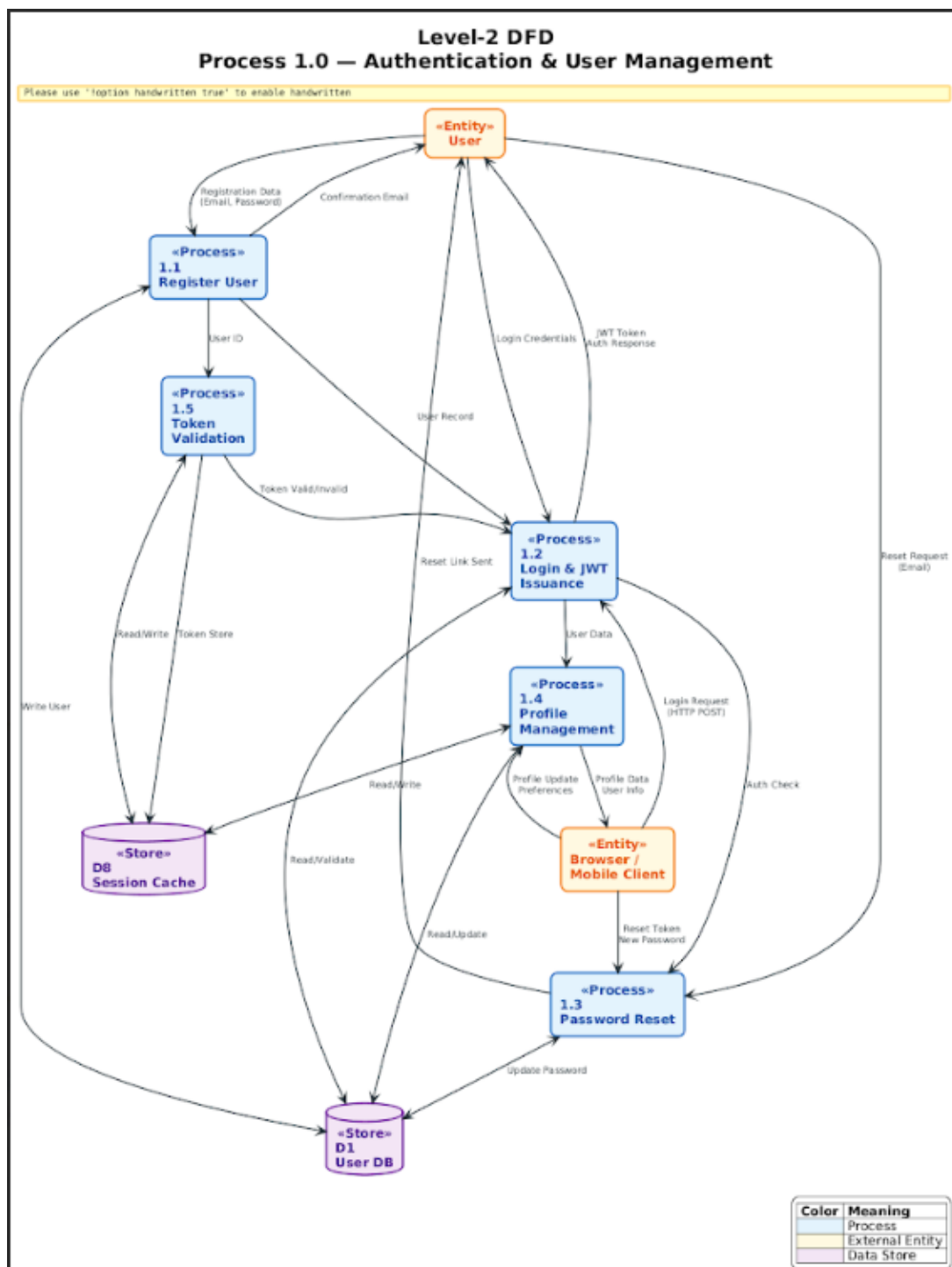


Figure 4: Level-2 DFD — Process 1.0: Authentication & User Management

2.3.3 Level-2 DFD — Process 2.0 (Detailed View): News Ingestion & Management

An alternative detailed view of the news ingestion pipeline, emphasizing the data quality control mechanisms and the duplicate detection workflow. This view highlights the feedback loops where validated articles are indexed while rejected articles are flagged for admin review. The diagram also shows the interaction between the Article Indexer and the Duplicate Detector to ensure content uniqueness before final storage in the News Article DB (D2).

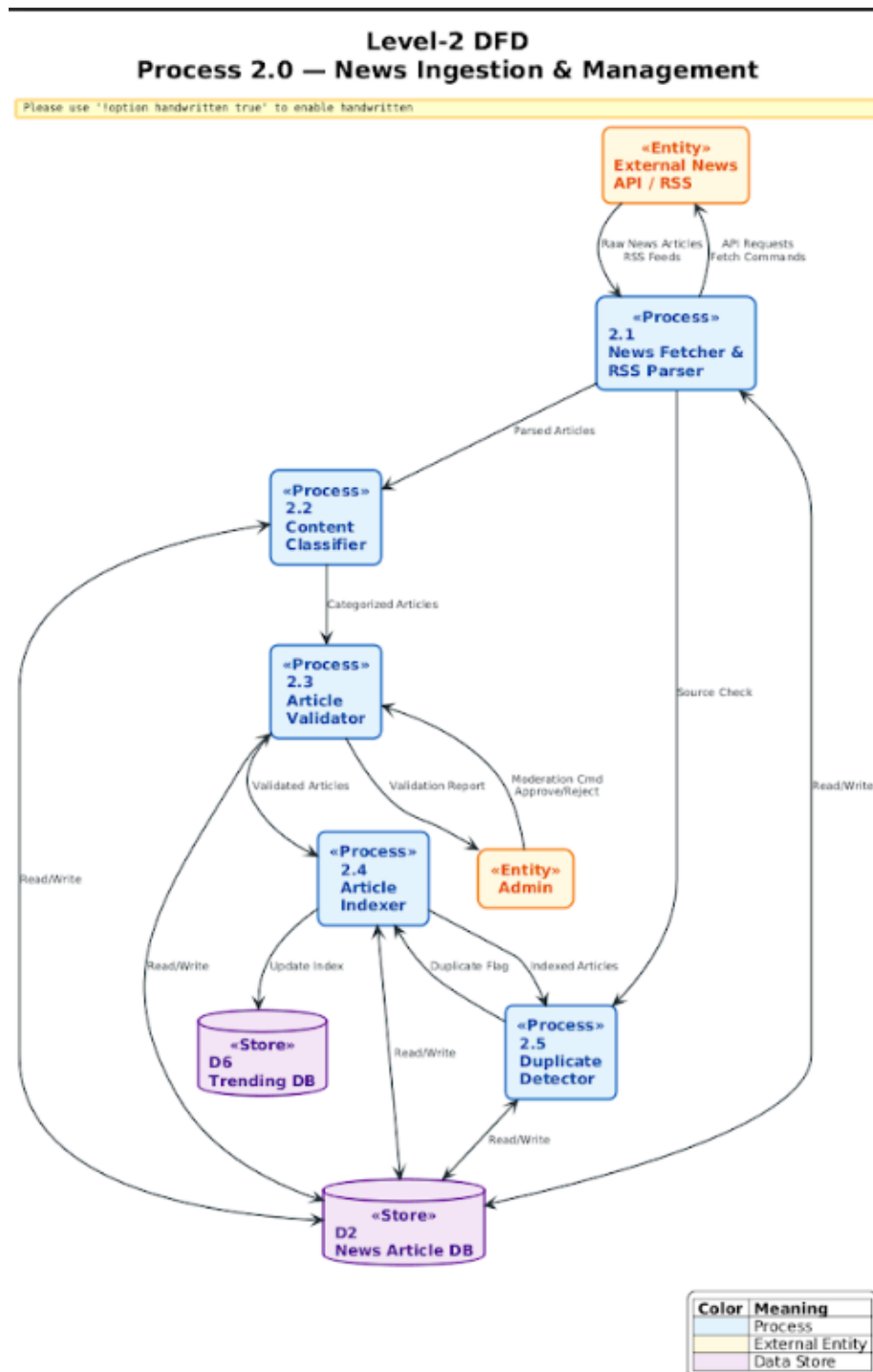


Figure 5: Level-2 DFD — Process 2.0 (Detailed View): News Ingestion & Management

2.3.4 Level-2 DFD — Process 3.0: Personalized Feed Engine

This diagram illustrates the recommendation and personalization pipeline that delivers tailored content to each user:

- **Process 3.1 — Category Browser:** Allows users to explore articles organized by categories (World, Technology, Sports, Entertainment, etc.). Retrieves categorized content from News Article DB (D2).
- **Process 3.2 — Search & Filter Engine:** Processes free-text search queries, applies advanced filters (date range, source, sentiment, reading time), and returns ranked results.
- **Process 3.3 — Feed Ranking & Personalization:** The core recommendation engine that ranks articles using collaborative filtering, content-based filtering, and hybrid ML algorithms. Considers user preferences (D3), reading history (D5), and bookmarked articles (D4).
- **Process 3.4 — Article Detail Renderer:** Formats and displays full article content with AI-generated summaries, related articles, and social sharing options.
- **Process 3.5 — Reading History Tracker:** Records user reading patterns including articles read, time spent, scroll depth, and engagement metrics. Updates Reading History (D5) for future personalization.
- **Process 3.6 — Favorites Manager:** Handles liked/saved articles, allowing users to curate personal collections.
- **Data Stores:** D1 (Preferences DB), D2 (News Articles DB), D4 (Bookmark DB), D5 (Reading History).
- **External Entities:** User/Browser for content consumption, Auth System for validation.

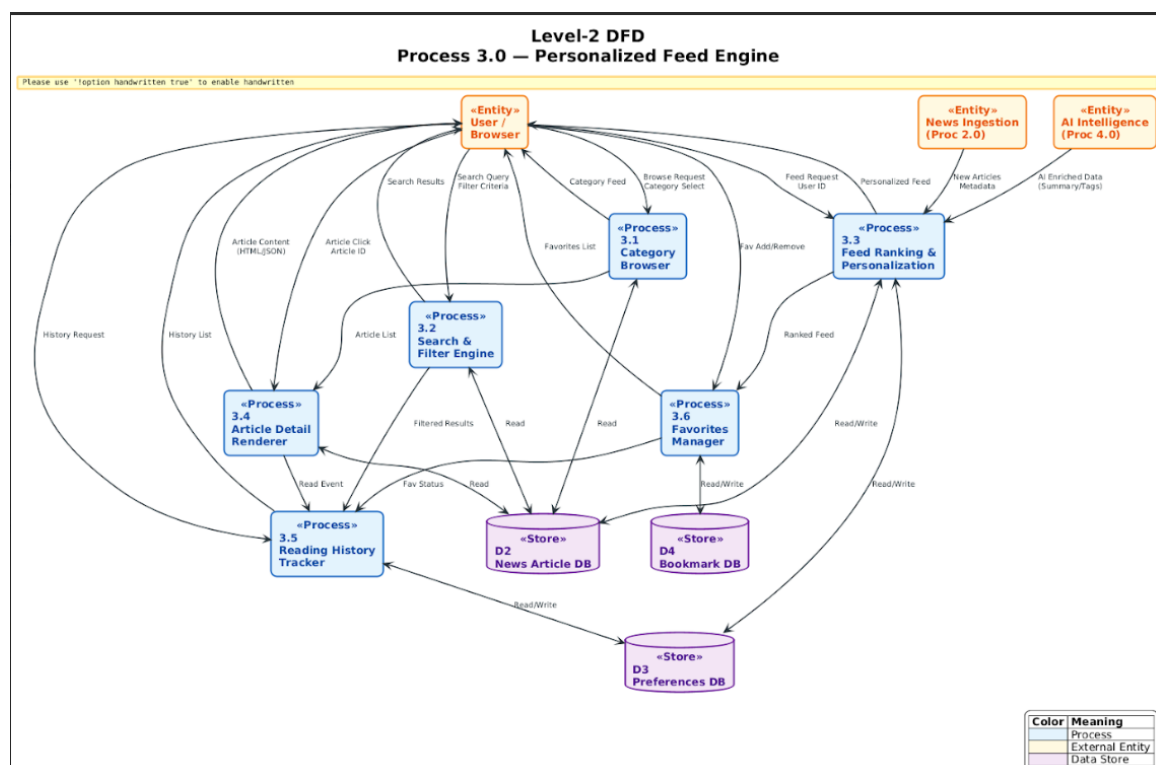


Figure 6: Level-2 DFD — Process 3.0: Personalized Feed Engine

2.3.5 Level-2 DFD — Process 4.0: AI Intelligence Processor

The AI processing pipeline that leverages Ollama AI (Llama 3.2) for intelligent content processing:

- **Process 4.1 — Planner & Request Decomposer:** Analyzes complex user queries and breaks them into sub-tasks. Determines which AI capabilities are needed (summarization, Q&A, sentiment analysis, topic extraction).
- **Process 4.2 — Executor & Tool Invoker:** Calls appropriate AI tools and services based on the planner's decomposition. Manages API calls to Ollama AI with proper prompt engineering.
- **Process 4.3 — Context & Memory Manager:** Maintains conversation context across multiple interactions, stores user-specific preferences for AI responses, and manages session state.
- **Process 4.4 — Response Formatter:** Structures AI output for user presentation, including markdown formatting, citation linking, and confidence scoring.
- **Data Stores:** D2 (News Article DB) for source content, AI Cache for performance optimization.
- **External Entity:** Ollama AI (Llama 3.2) for LLM operations and User/Browser for result display.

Level-2 DFD Process 4.0 — AI Intelligence Processor

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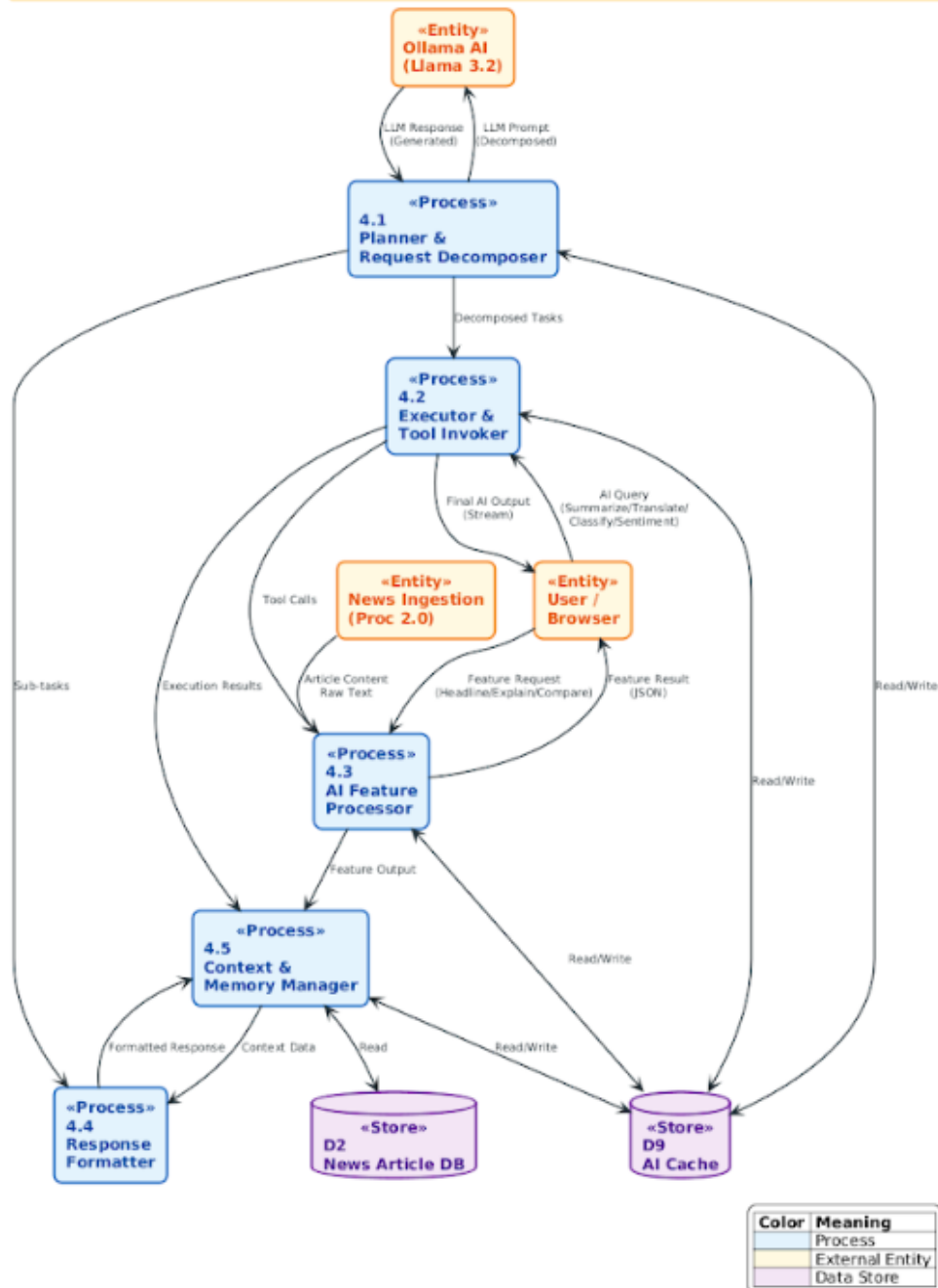


Figure 7: Level-2 DFD — Process 4.0: AI Intelligence Processor

2.3.6 Level-2 DFD — Process 5.0: Trending & Analytics

The analytics and trending detection system that identifies popular content and generates insights:

- **Process 5.1 — Topic Extractor:** Identifies key topics and themes from article metadata and content using NLP techniques. Extracts named entities, keywords, and semantic clusters.
- **Process 5.2 — Engagement Scorer:** Calculates comprehensive engagement metrics including views, shares, comments, time spent, click-through rates, and social media mentions.
- **Process 5.3 — Trending Ranker:** Ranks topics by popularity velocity (rate of growth), absolute popularity, and geographic distribution. Identifies emerging trends before they peak.
- **Process 5.4 — Analytics Dashboard:** Generates visual reports and KPIs for administrators, including user growth, content performance, engagement trends, and system health metrics.
- **Data Stores:** D2 (News Articles DB) for source data, D6 (Trending DB) for computed metrics.
- **External Entity:** User/Browser receives trending topic displays and popular article recommendations.

Level-2 DFD Process 5.0 — Trending & Analytics

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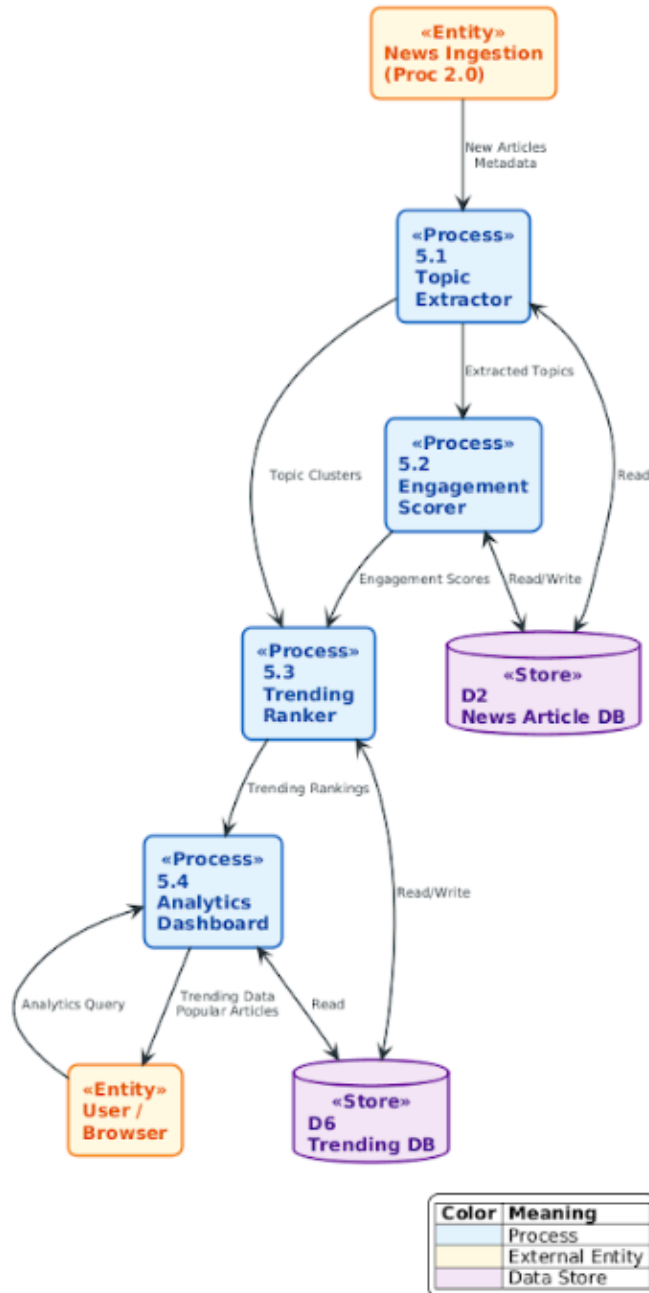


Figure 8: Level-2 DFD — Process 5.0: Trending & Analytics

2.3.7 Level-2 DFD — Process 6.0: Notification Manager

The notification delivery pipeline that keeps users informed through multiple channels:

- **Process 6.1 — Preference Manager:** Stores and retrieves user notification settings including channel preferences (email, SMS, push), frequency (immediate, daily digest, weekly digest), and content categories of interest.
- **Process 6.2 — Breaking News Detector:** Identifies high-priority news events by monitoring article velocity, source credibility, and topic significance. Triggers immediate alerts for critical events.
- **Process 6.3 — Digest Compiler:** Aggregates relevant articles into daily or weekly digest formats based on user preferences. Personalizes content selection using the Personalized Feed Engine.
- **Process 6.4 — Notification Dispatcher:** Routes notifications to appropriate delivery channels based on user preferences. Formats messages for each channel (HTML for email, SMS text, push notification payload).
- **Process 6.5 — Delivery Tracker:** Monitors notification delivery status, tracks open rates, click-through rates, and bounce rates. Updates delivery logs in Notification DB (D7).
- **Data Stores:** D3 (Preferences DB) for user settings, D7 (Notification DB) for delivery logs.
- **External Entity:** Email/SMS Gateway for actual message delivery.

Level-2 DFD Process 6.0 — Notification Manager

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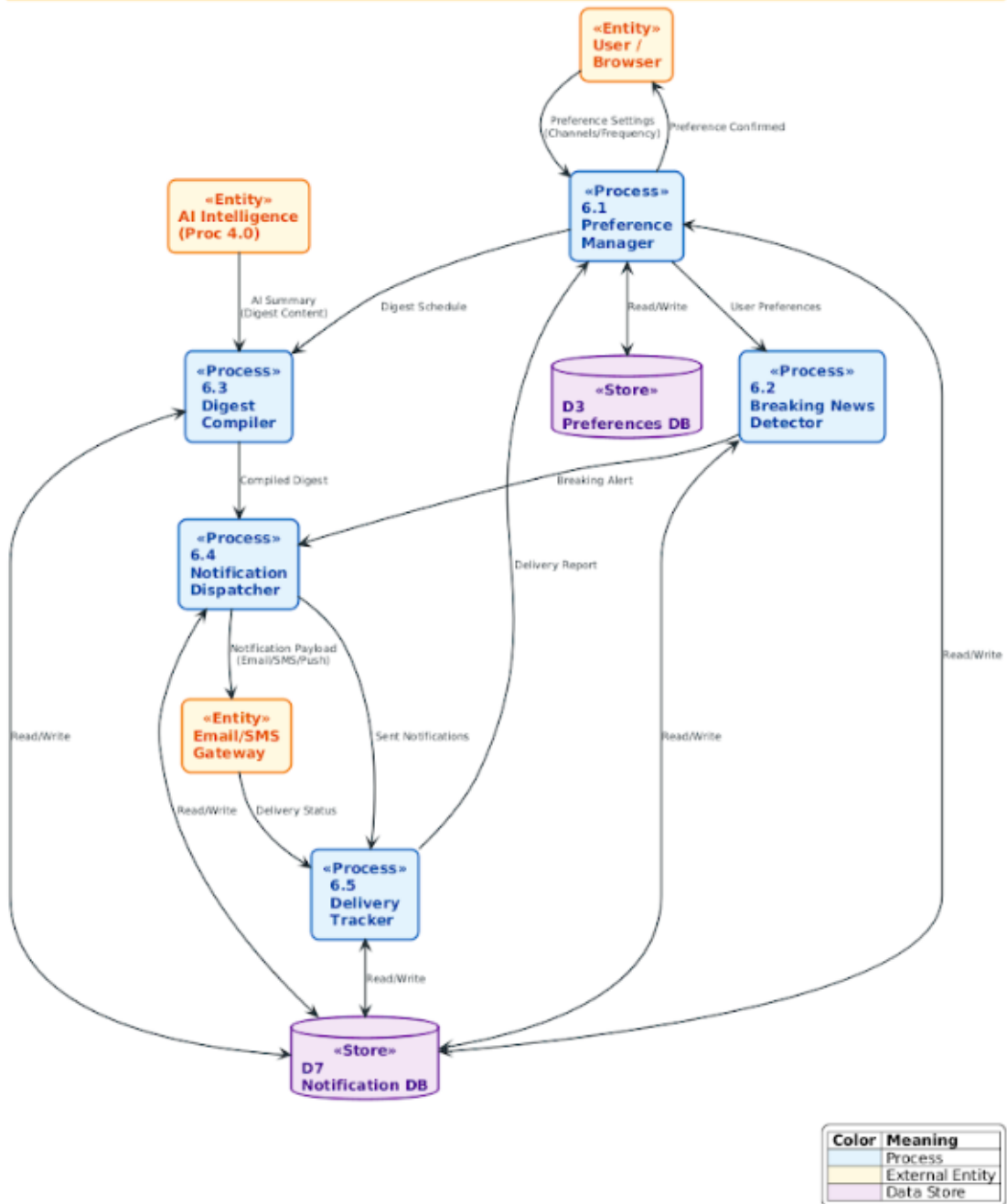


Figure 9: Level-2 DFD — Process 6.0: Notification Manager

2.3.8 Level-2 DFD — Process 7.0: Bookmark & Offline Manager

The bookmark and offline reading management system that enhances user experience across devices:

- **Process 7.1 — Bookmark Creator:** Saves articles to user bookmarks with metadata (timestamp, category, notes). Validates authentication before creating entries.
- **Process 7.2 — Bookmark Lister:** Retrieves and displays saved bookmarks with sorting options (date, category, source). Supports pagination and search within bookmarks.
- **Process 7.3 — Bookmark Remover:** Deletes bookmarks on user request with confirmation. Handles bulk deletion operations.
- **Process 7.4 — Offline Cache Manager:** Downloads article content for offline access, including text, images, and essential metadata. Manages storage quotas and cache expiration.
- **Process 7.5 — Sync Engine:** Synchronizes bookmark data across user devices using conflict resolution strategies (last-write-wins, merge). Ensures consistency between mobile and web clients.
- **Data Stores:** D2 (News Articles DB) for content, D4 (Bookmark DB) for user collections.
- **External Entity:** Auth System (Proc 1.0) for authentication validation.

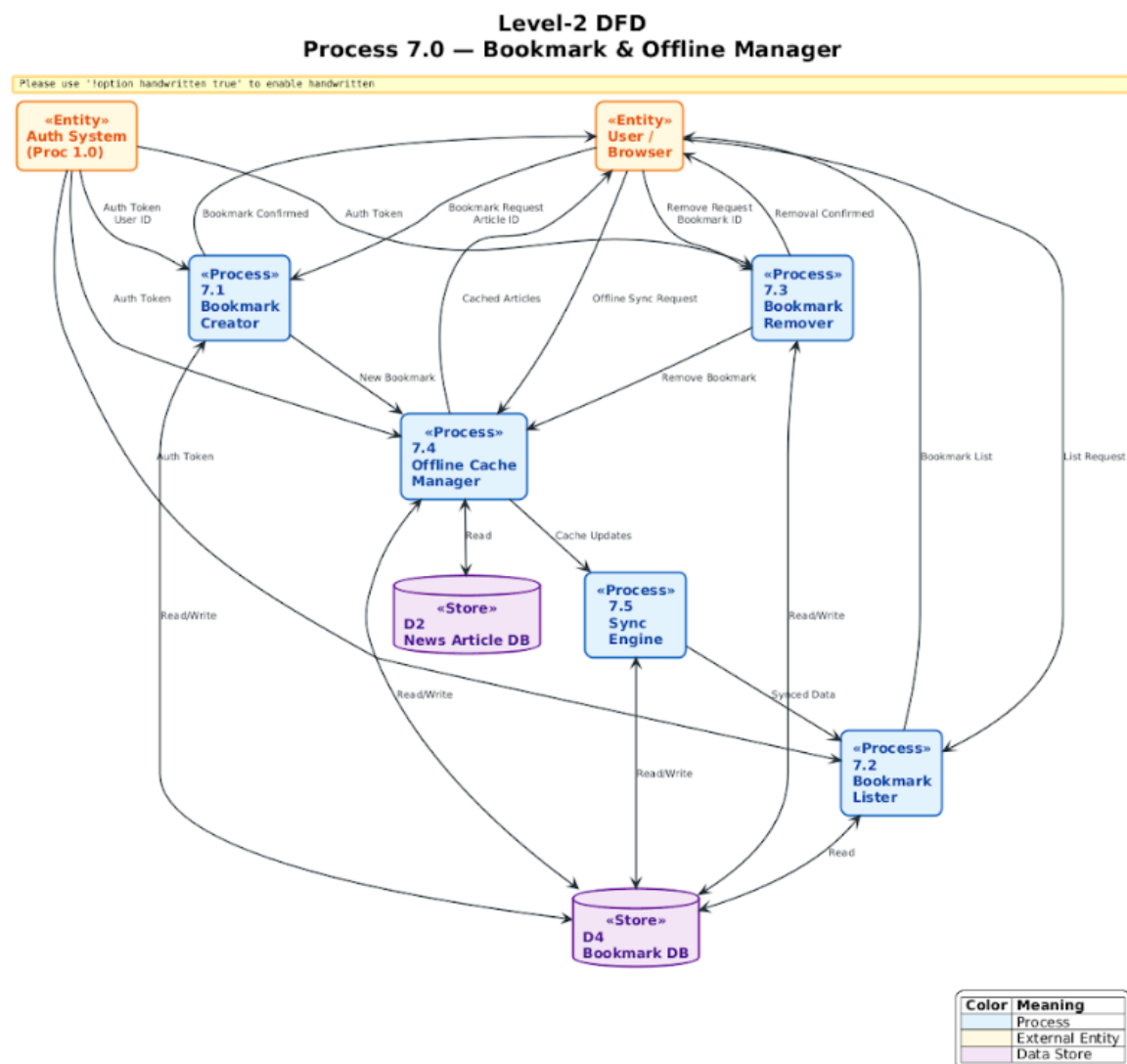


Figure 10: Level-2 DFD — Process 7.0: Bookmark & Offline Manager

2.3.9 Level-2 DFD — Process 8.0: Feedback & Admin Console

The administrative and feedback management system that ensures content quality and system health:

- **Process 8.1 — Feedback & Rating Collector:** Gathers user ratings (1-5 stars), written feedback, and content reports. Validates submissions for spam and stores in Feedback DB (D10).
- **Process 8.2 — Content Moderation:** Reviews user-generated content, flags inappropriate material, and approves/rejects submissions. Uses both automated filtering and manual admin review.
- **Process 8.3 — Bug Report Handler:** Processes bug reports from users, categorizes issues by severity and component, assigns to development teams, and tracks resolution status.
- **Process 8.4 — Admin Dashboard & Analytics:** Provides system-wide analytics including user activity, content performance, system errors, and AI accuracy metrics. Generates exportable reports.
- **Process 8.5 — User Management & RBAC:** Manages user roles (user, moderator, admin, super-admin) and access permissions. Handles account suspension, role upgrades, and permission audits.
- **Data Stores:** D1 (User DB) for account data, D8 (Admin Logs) for audit trails, D10 (Feedback DB).
- **External Entities:** Admin for management actions, Auth System (Proc 1.0) for validation, User/Browser for feedback submission.

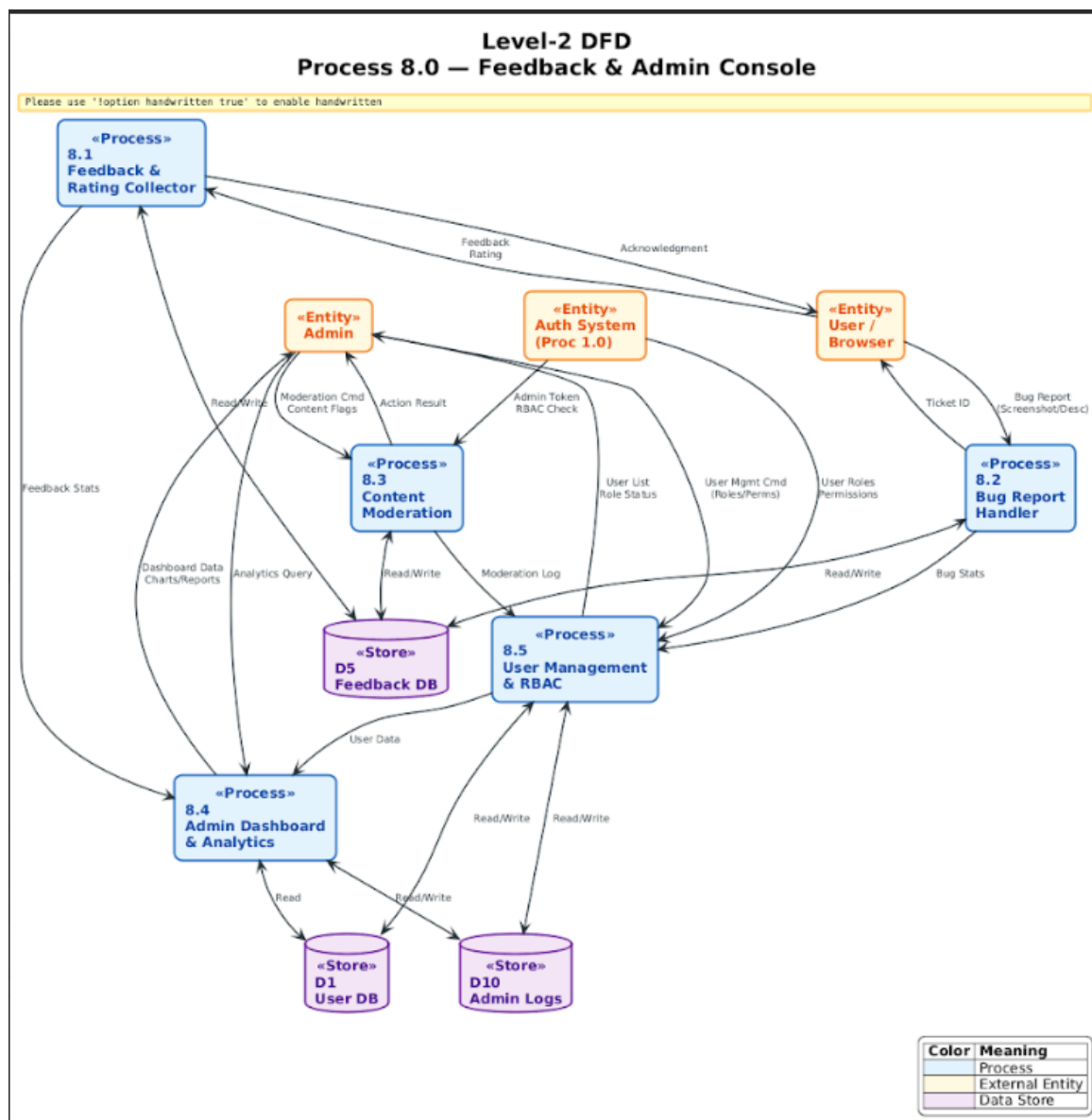


Figure 11: Level-2 DFD — Process 8.0: Feedback & Admin Console

3. UML Behavioral Diagrams

UML Behavioral Diagrams model the dynamic aspects of the system, showing how objects interact over time. This section covers the Sequence Diagram and Collaboration Diagram for the **World In Brief** platform, illustrating the temporal and structural dimensions of system interactions respectively.

3.1 Sequence Diagram

The **Sequence Diagram** illustrates the time-ordered interaction between objects in the system for a specific use case. It emphasizes the chronological sequence of messages exchanged between participants, represented as vertical lifelines with horizontal message arrows.

Key Participants (Lifelines):

- **User / Browser:** Initiates requests and receives responses. Represents the client-side presentation layer.
- **Auth Controller:** Manages authentication and session validation. Handles login, logout, token refresh, and permission checks.
- **Feed Controller:** Handles feed generation and article retrieval. Coordinates between multiple services to build personalized content.
- **AI Service:** Processes AI-related requests including summarization, Q&A, sentiment analysis, and topic extraction.
- **News Ingestion Service:** Manages article fetching, parsing, classification, and storage from external sources.
- **Notification Service:** Handles alert generation, digest compilation, and delivery coordination.
- **Database Layer:** Persistent storage for all entities including user data, articles, preferences, and analytics.
- **Ollama AI:** External LLM service for advanced natural language processing tasks.
- **Email/SMS Gateway:** External notification delivery service.

Typical Interaction Flow:

1. **Authentication Phase:** User sends login credentials → Auth Controller validates against User DB → issues JWT token → stores session in Session Cache.
2. **Feed Request Phase:** User requests personalized feed → Feed Controller queries Preferences DB and Reading History → fetches ranked articles from News Article DB → applies AI summarization if needed.
3. **AI Processing Phase:** For complex queries, Feed Controller calls AI Service → AI Service queries Ollama AI → receives structured response → caches result in AI Cache → returns to Feed Controller.
4. **Bookmark Phase:** User bookmarks an article → Bookmark Service validates auth → updates Bookmark DB (D4) → confirms action to user.
5. **Notification Phase:** Breaking news triggers Notification Service → checks user preferences → compiles alert → dispatches via Email/SMS Gateway → logs delivery in Notification DB (D7).
6. **Admin Phase:** Admin actions flow through Admin Console → respective services → updates Admin Logs (D8) for audit trails.

Significance: Sequence diagrams are critical for understanding temporal dependencies, identifying potential bottlenecks in message flows, and designing precise API contracts between services. They help developers understand the exact order of operations and where synchronous vs. asynchronous calls should be used.

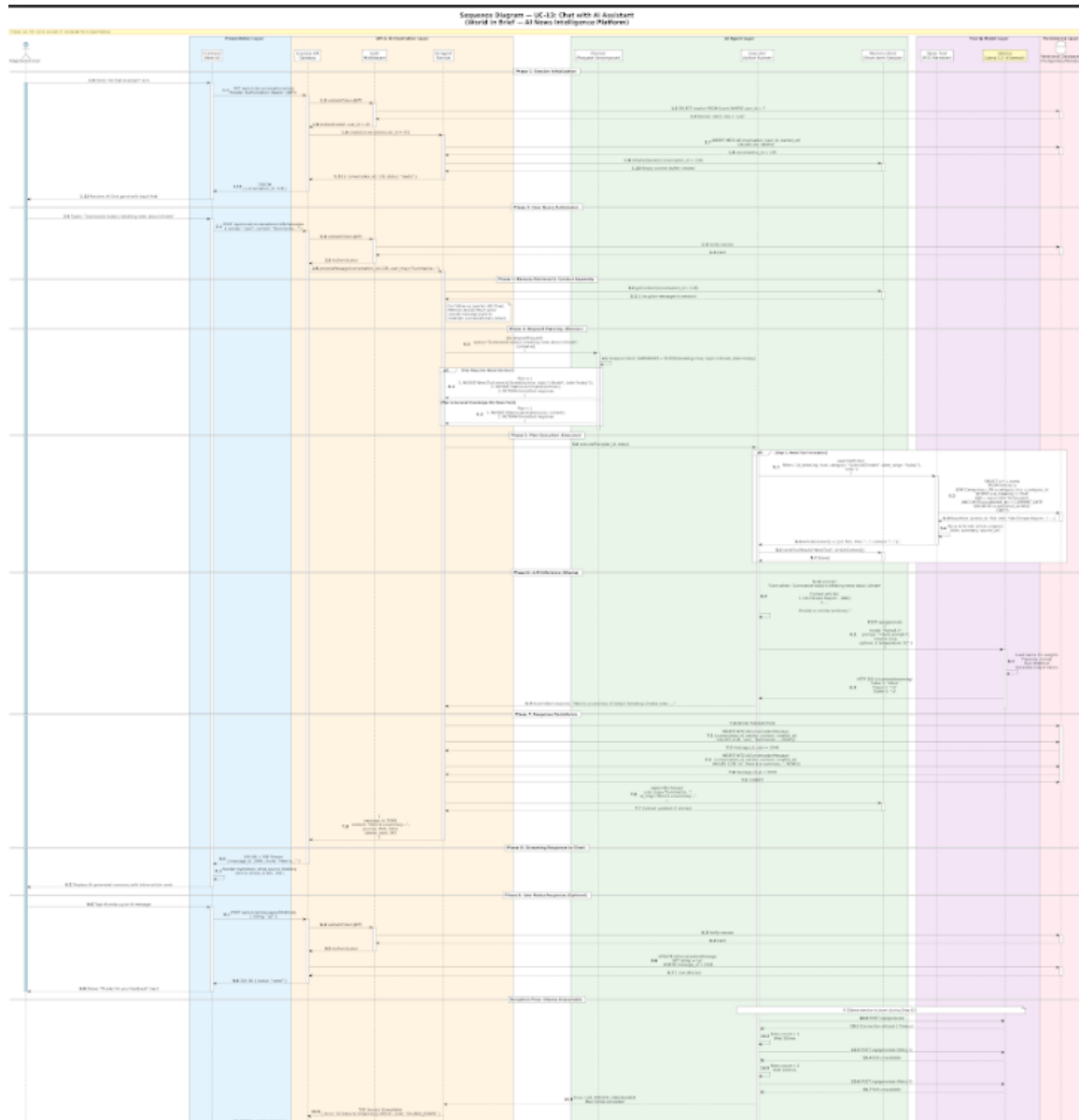


Figure 12: Sequence Diagram — Time-ordered Interactions in World In Brief Platform

3.2 Collaboration Diagram

The **Collaboration Diagram** (also known as a Communication Diagram) shows the structural organization of objects that send and receive messages. Unlike the Sequence Diagram, it emphasizes the relationships and links between objects rather than the strict time ordering of messages.

Key Characteristics:

- Objects are represented as nodes in a graph-like structure, showing their structural relationships.
- Links (edges) between objects represent communication pathways and associations.
- Numbered messages indicate the sequence of interactions while preserving structural context.
- Focuses on the *context* of interactions — which objects are connected and how they collaborate.

Objects and Their Links:

- **UserInterface** connects to **AuthManager** (login/logout), **FeedEngine** (content requests), **BookmarkManager** (save/retrieve bookmarks), and **NotificationManager** (alert preferences).
- **FeedEngine** links to **NewsDB** (article retrieval), **PreferenceDB** (personalization data), **AIProcessor** (summarization requests), and **AnalyticsEngine** (engagement tracking).
- **AIProcessor** connects to **OllamaAI** (LLM API calls) and **AICache** (response caching for performance).
- **NotificationManager** links to **UserDB** (preference retrieval) and **GatewayService** (delivery dispatch).
- **AdminConsole** connects to **ModerationService** (content review), **AnalyticsEngine** (dashboard data), and **UserManagementService** (RBAC operations).
- **NewsIngestionService** links to **ExternalNewsAPI** (content fetching) and **NewsDB** (storage).

Comparison with Sequence Diagram:

While both diagrams model the same interactions, the Collaboration Diagram provides a better structural overview of object relationships, making it ideal for understanding system topology and object coupling. The Sequence Diagram is preferred for analyzing timing, concurrency, and performance bottlenecks. Together, they provide a complete picture: the Collaboration Diagram shows *who talks to whom*, while the Sequence Diagram shows *when they talk*.

Significance: Collaboration diagrams help architects understand object dependencies, design patterns (such as Observer, Facade, and Mediator), and the overall structural coherence of the system's runtime behavior. They are particularly valuable during refactoring and when assessing the impact of architectural changes.

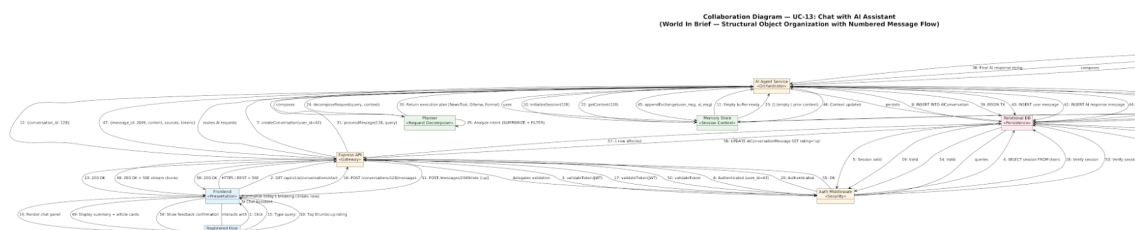


Figure 13: Collaboration Diagram — Object Interactions in World In Brief Platform

4. Conclusion

This document has presented a comprehensive system design for the **World In Brief — AI News Intelligence Platform** using structured analysis and UML behavioral modeling techniques.

The **Data Flow Diagrams** (Levels 0, 1, and 2) progressively decomposed the system from a high-level context view down to detailed process specifications. Level-0 established the system boundary and external interactions, identifying six external entities and their data exchanges. Level-1 revealed the eight major functional modules and ten data stores that form the backbone of the platform. Level-2 provided granular insights into each module's internal operations, data transformations, validation logic, and storage requirements.

The **Sequence Diagram** captured the temporal dynamics of object interactions, essential for API design, performance analysis, and understanding synchronous/asynchronous communication patterns. The **Collaboration Diagram** complemented this by emphasizing the structural relationships between objects, aiding in architectural decisions regarding coupling, cohesion, and design pattern application.

Together, these diagrams provide a complete blueprint for developers, testers, and stakeholders to understand, implement, and maintain the **World In Brief** platform. The modular design ensures scalability and maintainability, the AI integration enables intelligent and personalized content delivery, and the multi-channel notification system guarantees timely user engagement. The comprehensive data store architecture supports efficient querying, while the external entity integrations ensure seamless connectivity with third-party services.

This documentation serves as a foundation for agile development, allowing teams to work on individual processes while maintaining a holistic understanding of system interactions. The separation of concerns demonstrated in the DFDs aligns with modern microservices architecture principles, making the system ready for cloud-native deployment and continuous integration pipelines.